Nutrition and Dietary Guidelines
Balanced Diet Basics
A balanced diet provides the body with essential nutrients needed for proper function, growth, and disease prevention. According to the World Health Organization and USDA Dietary Guidelines, a healthy diet includes:
Macronutrients
Carbohydrates (45-65% of daily calories)
- Primary energy source for the body
- Choose complex carbohydrates: whole grains, oats, brown rice, quinoa
- Limit refined carbohydrates: white bread, sugary cereals, pastries
- Recommended fiber intake: 25-30 grams per day
Proteins (10-35% of daily calories)
- Essential for muscle repair, immune function, and enzyme production
- Sources: lean meats, fish, eggs, legumes, nuts, tofu, dairy
- Recommended: 0.8 grams per kilogram of body weight for adults
- Higher needs during pregnancy, illness, or intense physical activity
Fats (20-35% of daily calories)
- Necessary for hormone production, nutrient absorption, and cell structure
- Healthy fats: olive oil, avocados, nuts, seeds, fatty fish (omega-3)
- Limit saturated fats: less than 10% of daily calories
- Avoid trans fats: found in processed and fried foods
Micronutrients
Essential Vitamins
- Vitamin A: Vision, immune function (carrots, sweet potatoes, spinach)
- Vitamin C: Immune support, wound healing (citrus fruits, bell peppers, strawberries)
- Vitamin D: Bone health, immune function (sunlight, fortified foods, fatty fish)
- Vitamin B12: Nerve function, red blood cell formation (meat, eggs, dairy)
- Folate: Cell division, especially important during pregnancy (leafy greens, legumes)
Essential Minerals
- Calcium: Bone and teeth health (dairy, leafy greens, fortified foods) · 1000mg/day for adults
- Iron: Oxygen transport in blood (red meat, spinach, legumes) · 8mg/day for men, 18mg/day for premenopausal women
- Potassium: Heart and muscle function (bananas, potatoes, beans) · 2600-3400mg/day
- Zinc: Immune function, wound healing (meat, shellfish, legumes) · 8-11mg/day
Source: USDA Dietary Guidelines 2020-2025, WHO
Hydration
Water is essential for virtually every body function including temperature regulation, nutrient transport, and waste removal.
Daily Water Intake Recommendations
- Adult men: approximately 3.7 liters (about 13 cups) per day
- Adult women: approximately 2.7 liters (about 9 cups) per day
- Includes water from all beverages and food (about 20% comes from food)
Signs of Dehydration
- Thirst and dry mouth
- Dark yellow urine
- Fatigue and dizziness
- Decreased urine output
- Headache
Tips for Staying Hydrated
- Carry a reusable water bottle
- Drink water before, during, and after exercise
- Eat water-rich foods (watermelon, cucumber, oranges)
- Set reminders to drink water throughout the day
- Limit sugary drinks and excessive caffeine
Source: National Academy of Sciences, WHO
Special Dietary Considerations
Heart-Healthy Diet

- Emphasize fruits, vegetables, whole grains, and lean proteins
- Choose healthy fats (olive oil, nuts, avocados)
- Limit sodium to less than 2,300mg per day
- Limit added sugars to less than 10% of daily calories
- Include fatty fish at least twice a week for omega-3 fatty acids

Diabetes-Friendly Eating

- Monitor carbohydrate intake and choose complex carbs
- Eat at regular intervals to maintain stable blood sugar
- Include fiber-rich foods to slow glucose absorption
- Limit sugary beverages and refined carbohydrates
- Work with a healthcare provider for personalized meal planning

Vegetarian and Vegan Nutrition

- Ensure adequate protein from varied plant sources
- Consider vitamin B12 supplementation (especially for vegans)
- Include iron-rich plant foods with vitamin C for better absorption
- Ensure adequate calcium through fortified foods or supplements
- Include omega-3 sources: flaxseed, chia seeds, walnuts

Source: American Heart Association, American Diabetes Association, NIH

Foods to Limit

The following should be consumed in moderation for optimal health:

- **Added sugars**: Limit to less than 25g/day for women, 36g/day for men
- **Sodium**: Limit to less than 2,300mg/day (about 1 teaspoon of salt)
- **Processed meats**: Associated with increased risk of certain cancers and heart disease
- **Ultra-processed foods**: Often high in unhealthy fats, sugars, and sodium
- **Alcohol**: If consumed, limit to 1 drink/day for women, 2 drinks/day for men

Source: WHO, American Heart Association, CDC